

Eddy Ndumia

AI enthusiast & data engineer (AWS) - Mathematics & Computer Science, JKUAT

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SUMMARY

AI enthusiast with data engineering experience on AWS, turning raw, messy data into pipelines and models that hold up in production. Currently building PesaScore, a fintech platform that scores creditworthiness from alternative data (M-Pesa statements) instead of a traditional credit bureau, and Capstone, an AI lead assistant for real estate agents. Freelance web developer since 2020.

EXPERIENCE

Independent Builder - PesaScore, Capstone, Chukua & Kwetu

2026 - Present

- PesaScore: alt-data credit scoring from M-Pesa statements. Parser tested on a real 7,400-transaction statement, explainable 300-850 scorecard, borrower consent, Postgres RLS. Flutter app, FastAPI, lender web portal.
- Capstone: AI agent handling WhatsApp/Instagram DMs for real estate agencies - qualifies leads, books viewings, hands off to a human when it should. FastAPI + Postgres backend, Next.js dashboard.
- Chukua: Flutter app for giving away free stuff nearby, with AI value estimates and M-Pesa claim fees. Supabase + PostGIS, atomic claim holds, Row-Level Security keeping givers' phones private.
- Kwetu: Flutter app for people abroad to buy electronics for family in Kenya, paid in USDT/USDC.

Freelance Web Developer - Kenya

2020 - Present

- Responsive, data-driven sites and custom full-stack builds for small businesses - WordPress and code-first, depending on the client.

Project Lead, Web Housing Management System - JKUAT

Sept - Dec 2022

- Led a university team building a cloud-backed system for tenants, properties, and admin tasks - auth, data tracking, reporting dashboards.

EDUCATION & TRAINING

B.Sc. Mathematics and Computer Science - JKUAT

2020 - 2026 (graduating December 2026)

ALX / Holberton Software Engineering Program

2023 - low-level programming in C, algorithms, Linux/DevOps

HackRU IX Hackathon

2022 - patient-connection platform for the NY-Presbyterian/Morgan Stanley Children's Hospital Challenge

Certifications

Data Science with Python (JKUAT) | Cisco Networking Level 1 (JKUAT) | Web & Mobile Application Development (eMobilis Technology Training Institute)

SELECTED PROJECTS

Sales ETL Pipeline - Messy raw sales data cleaned, tested, and loaded into a quality-gated SQLite warehouse; 6 automated data-quality checks gate every run.

Customer Churn Classification - Imbalance-aware model comparison (class weighting vs. SMOTE across 3 model types) translated into an estimated \$112k business impact, not just an accuracy score.

Sales Pipeline Funnel Analytics - 6 SQL queries answering real sales-ops questions: funnel drop-off, rep and lead-source win rates, deal velocity by segment, stage-weighted revenue forecasting.

Sales Demand Forecasting - Seasonal-naive vs. SARIMA vs. XGBoost on a 90-day holdout; found and explained why single-seasonality SARIMA misses yearly retail spikes XGBoost catches via calendar features.

Lead Scoring App - Gradient-boosting lead scorer deployed as a Streamlit app, with precision/recall thresholds tied to real sales team capacity (~1.9x lift over random contact at 25% capacity).

PropFire - Trading timer and journal for forex traders timing entries around high-impact news releases.

RAG Chatbot - LangChain-based retrieval-augmented chatbot for querying custom text datasets.

Full write-ups and repo links: ndumia.netlify.app/projects

SKILLS

Languages: Python, TypeScript, JavaScript/Node.js, Dart, Kotlin, SQL

Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, statsmodels, TensorFlow, LangChain / RAG

Infra: Postgres, PostGIS, Supabase, Docker, AWS

Web & mobile: React, Next.js, Flutter, FastAPI, Streamlit, WordPress